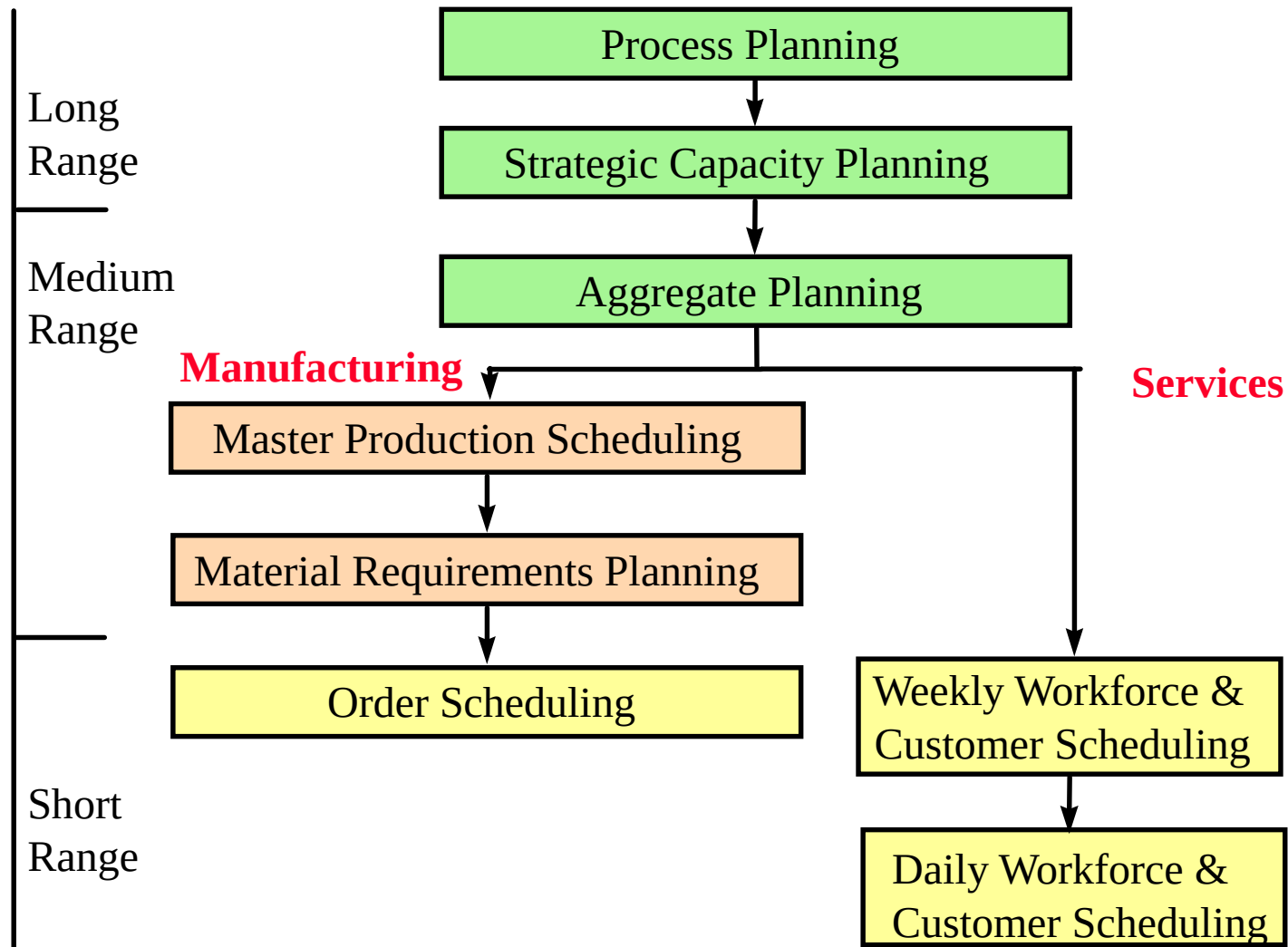


Operations Scheduling

Production Planning Process



Scheduling and Sequencing

- **Scheduling** is defined as the process of rank ordering the jobs in front of each resource with a view to maximize some chosen performance measure
- **Sequencing** is the ordering of the operations of the jobs in the operating system

Work Center

- An area in which production resources are organized and work is completed.
- May be a single machine, a group of machines, or an area where work is done.
- Can be organized according to function, product in a flow, or group technology
- Jobs need to be routed between functionally organized work centers to complete the work

Planning Context in the short term

- How do we assign the jobs to various work centers?
- Within each work center, how do we rank order the jobs?
- How do we assign other resources such as skilled workers and material handling devices to the operating system?
- How do we measure the performance of the operating system?

Objectives in Scheduling

- Meet due dates
- Minimize lead time
- Minimize setup time or cost
- Minimize work-in-process inventory
- Maximize machine utilization

It is unlikely, and often undesirable, to simultaneously satisfy all of these objectives

Scheduling Context

- **Number of jobs (n)**
- **Number of machines (m)**
- **Shop Configuration**
 - Job Shop
 - Flow shop
 - Cellular Manufacturing System
- **Job Priorities**
 - FCFS, SPT, LCFS, EDD, STR, CR, Random
- **Performance Measures**
 - Due date based: lateness, tardiness
 - Completion based: Flow time, make span

Priority Rules for Job Sequencing

1. **FCFS** (first come, first served). Orders are run in the order they arrive in the department.
2. **SOT** (shortest operating time). Run the job with the shortest completion time first, next-shortest second, and so on. This is sometimes also referred to as SPT (shortest processing time). This rule is often combined with a lateness rule to prevent jobs with longer times from being delayed too long.
3. **EDD** (earliest due date first). Run the job with the earliest due date first.
4. **STR** (slack time remaining). This is calculated as the time remaining before the due date minus the processing time remaining. Orders with the shortest slack time remaining (STR) are run first.

$$\text{STR} = \text{Time remaining before due date} - \text{Remaining processing time}$$

5. **STR/OP** (slack time remaining per operation). Orders with the shortest slack time per number of operations are run first.

$$\text{STR/OP} = \text{STR} / \text{Number of remaining operations}$$

Priority Rules for Job Sequencing

6. **LCFS** (last come, first served). This rule occurs frequently by default. As orders arrive, they are placed on the top of the stack; the operator usually picks up the order on top to run first.
7. **Random** order or whim. The supervisors or the operators usually select whichever job they feel like running.
8. **CR** (critical ratio). Critical ratio estimates the criticality of the job by computing a simple ratio using processing time information and due date. A smaller value of CR indicates that the job is more critical.

$$\text{Critical Ratio (CR)} = \frac{\text{Re maining time}}{\text{Re maining Work}} = \frac{(\text{Due Date} - \text{Current Date})}{\text{Re maining Pr oces sin g Time}}$$

If $CR > 1$, job ahead of schedule

If $CR < 1$, job behind schedule

If $CR = 1$, job on schedule

Performance Criterion

- **Flow time** is defined as the elapsed time between releasing a job into the shop and the time of completion of processing of the job

Release time of the job : R_i

Completion time of the job : C_i

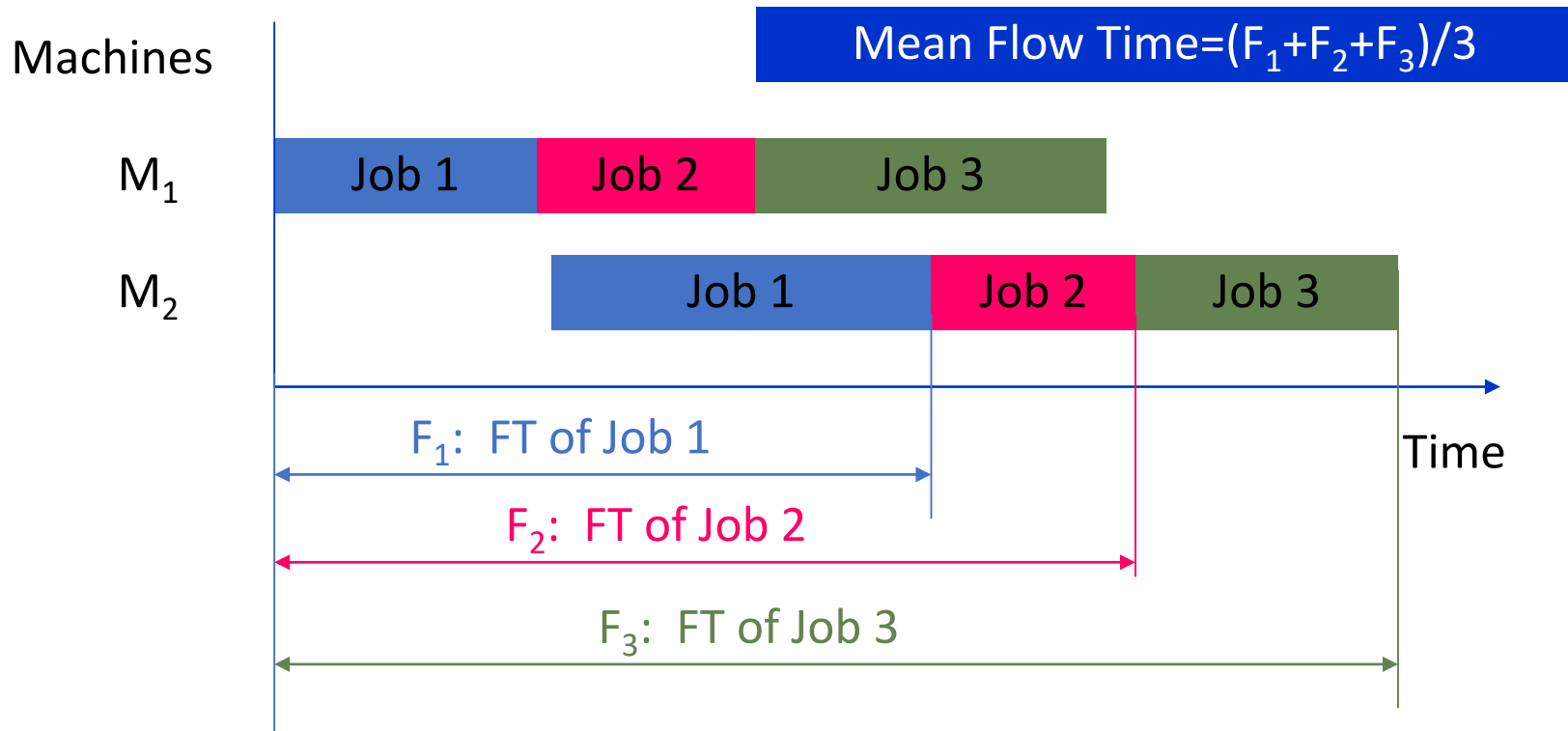
Flow time of the job : $F_i = (C_i - R_i)$

- **Make span** is defined as the time taken to complete all the jobs released into the shop for processing
Make span (Max. Completion time):

$$C_{\max} = \max_i \{C_i\}$$

Job Shop Scheduling Terminology

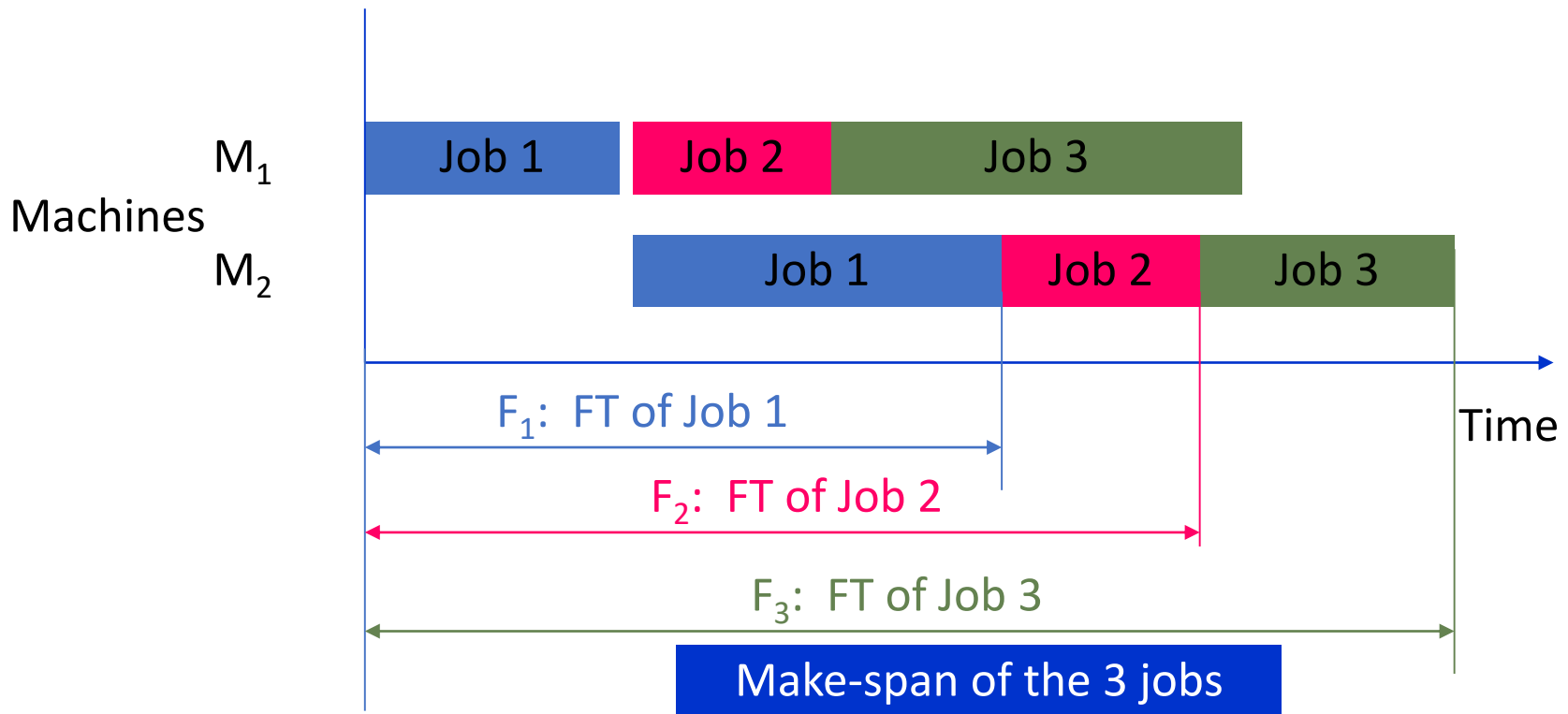
- Flow time
 - It is the amount of time job j spends in the system, that is, the time that elapses from the initiation of that job on the first machine to its completion.
 - The Mean Flow Time can be calculated as the arithmetic average of flow times for all n jobs.



Job Shop Scheduling Terminology

- Make Span

- The Make Span is the time required to complete all n jobs.
- Minimizing Make Span is one of the common objectives in multiple-machine sequencing problems.



Performance Criterion

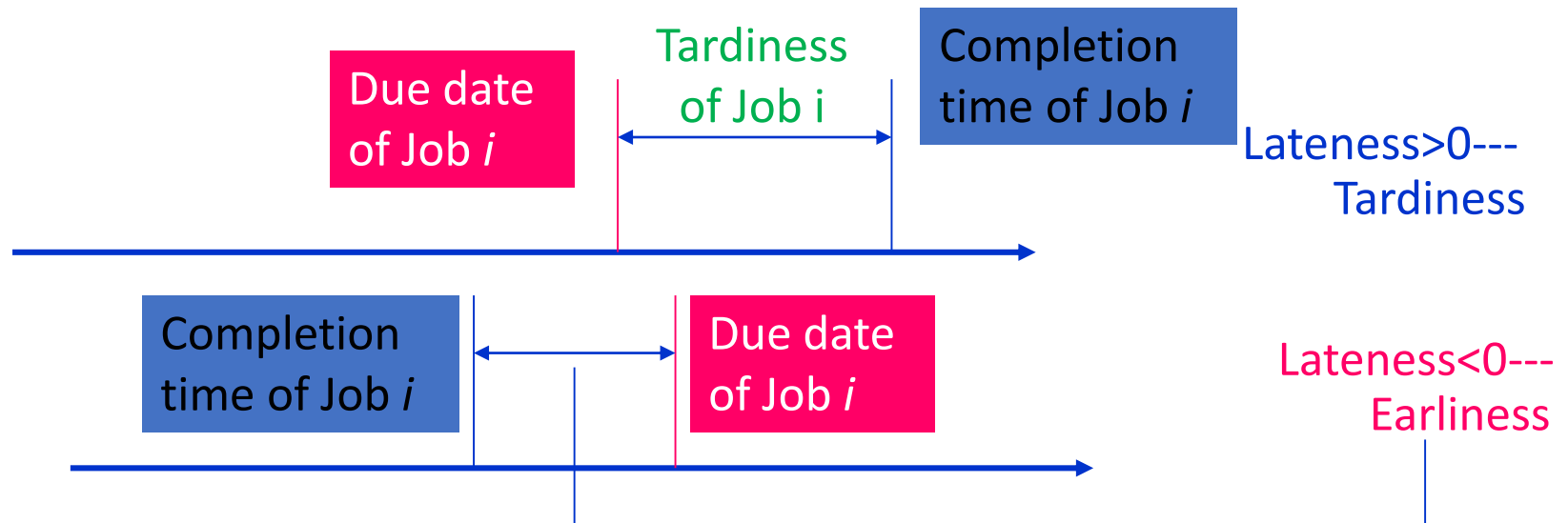
Due date based measures

- **Lateness** defined as the difference between completion time and due date.
If the due date for a job i is denoted as D_i , then
Lateness of the job: $L_i = (C_i - D_i)$
- If a job is completed ahead of time, instead of computing a negative value for L_i if we take zero, then the resulting measure is known as **tardiness**
Tardiness of the job: $T_i = \max(0, L_i)$

Job Shop Scheduling Terminology

- **Tardiness and Lateness**

- Tardiness is understood as positive difference between the Completion Time and the Due Date of a Job.
- Lateness also refers to the difference between the Completion Time and the Due Date of a Job but differs from tardiness in the sense that lateness can be either positive or negative.
- If lateness is positive, it is tardiness; when it is negative, it is earliness.



Where the Completion Time of a Job is earlier than its Due Date, the tardiness is 0.

Scheduling Solutions

- In Order to begin to attempt to develop a solution, break the problem into categories:
 - n jobs, 1 machine
 - n jobs, 2 machines
 - n jobs, 3 machines
 - n jobs, m machines



Example of Scenario 1: n Jobs, 1 Machine

A machine center in a job shop for a local fabrication company has five unprocessed jobs remaining at a particular point in time. The jobs are labeled 1, 2, 3, 4, and 5 in the order that they entered the shop. The respective processing times and due dates are given in the table below.

Sequence the 5 jobs and compare results based on mean flow time, average tardiness, and number of tardy jobs

Job number	Processing Time	Due Date
1	11	61
2	29	45
3	31	31
4	1	33
5	2	32

Scheduling Rules

FCFS

Mean Flow time= $268/5=53.6$
 Average tardiness= $121/5=24.2$
 No. of tardy jobs=3.

Job number	Processing Time	Due Date
1	11	61
2	29	45
3	31	31
4	1	33
5	2	32

Job	Completion Time	Due Date	Tardiness
1	11	61	0
2	40	45	0
3	71	31	40
4	72	33	39
5	74	32	42
Totals	268		121

Scheduling Rules

SPT

Mean Flow time= $135/5=27.0$

Average tardiness= $43/5=8.6$

No. of tardy jobs=1.

Job number	Processing Time	Due Date
1	11	61
2	29	45
3	31	31
4	1	33
5	2	32

Job	Processing Time	Completion Time	Due Date	Tardiness
4	1	1	33	0
5	2	3	32	0
1	11	14	61	0
2	29	43	45	0
3	31	74	31	43
Totals		135		43

Scheduling Rules

EDD

Mean Flow time= $235/5=47.0$
 Average tardiness= $33/5=6.6$
 No. of tardy jobs=4.

Job number	Processing Time	Due Date
1	11	61
2	29	45
3	31	31
4	1	33
5	2	32

Job	Processing Time	Completion Time	Due Date	Tardiness
3	31	31	31	0
5	2	33	32	1
4	1	34	33	1
2	29	63	45	18
1	11	74	61	13
Totals		235		33

Scheduling Rules

CR

Current time: $t=0$

Job number	Processing Time	Due Date	Critical Ratio
1	11	61	61/11(5.545)
2	29	45	45/29(1.552)
3	31	31	31/31(1.000)
4	1	33	33/1 (33.00)
5	2	32	32/2 (16.00)

Current time should be reset after scheduling one job

Current time: $t=31$

Job number	Processing Time	Due Date-Current Time	Critical Ratio
1	11	30	30/11(2.727)
2	29	14	14/29(0.483)
4	1	2	2/1 (2.000)
5	2	1	1/2 (0.500)

Scheduling Rules

CR

Mean Flow time= $289/5=57.8$

Average tardiness= $87/5=17.4$

No. of tardy jobs=4.

Current time=60

Job number	Processing Time	Due Date- Current Time	Critical Ratio
1	11	1	$1/11(0.0909)$
4	1	-27	$-27/1 < 0$
5	2	-28	$-28/2 < 0$

Both Jobs 4 and 5 are later, however Job 4 has shorter processing time and thus is scheduled first; Finally, job 1 is scheduled last.

Job number	Processing Time	Completion Time	Due Date	Tardiness
3	31	31	31	0
2	29	60	45	15
4	1	61	33	28
5	2	63	32	31
1	11	74	61	13
Totals		289		87

Scheduling Rules

SUMMARY

Rule	Mean Flow Time	Average Tardiness	Number of Tardy Jobs
FCFS	53.6	24.2	3
SPT	27.0	8.6	1
EDD	47.0	6.6	4
CR	57.8	17.4	4

Discussions

- SPT results in smallest mean flow time;
- EDD yields the minimum average;

Scenario 2: n jobs, 2 machines, flow shop

- Two or more jobs must be processed on two machines in a common sequence
- Wish to minimize the flow time from the beginning of the first job to the finish of the last job
- Use *Johnson's rule*
 1. List the operation time for each job
 2. Select the shortest operation time
 3. If the shortest time is on the first machine, do first
 - If the shortest time is on the second machine, do the job last
 - For ties, do first
 4. Repeat steps 2-3 for each remaining job

Johnson's Rule Conditions

- Job time must be known and constant
- Job times must be independent of sequence
- Jobs must follow same two-step sequence
- Job priorities cannot be used
- All units must be completed at the first work center before moving to the second

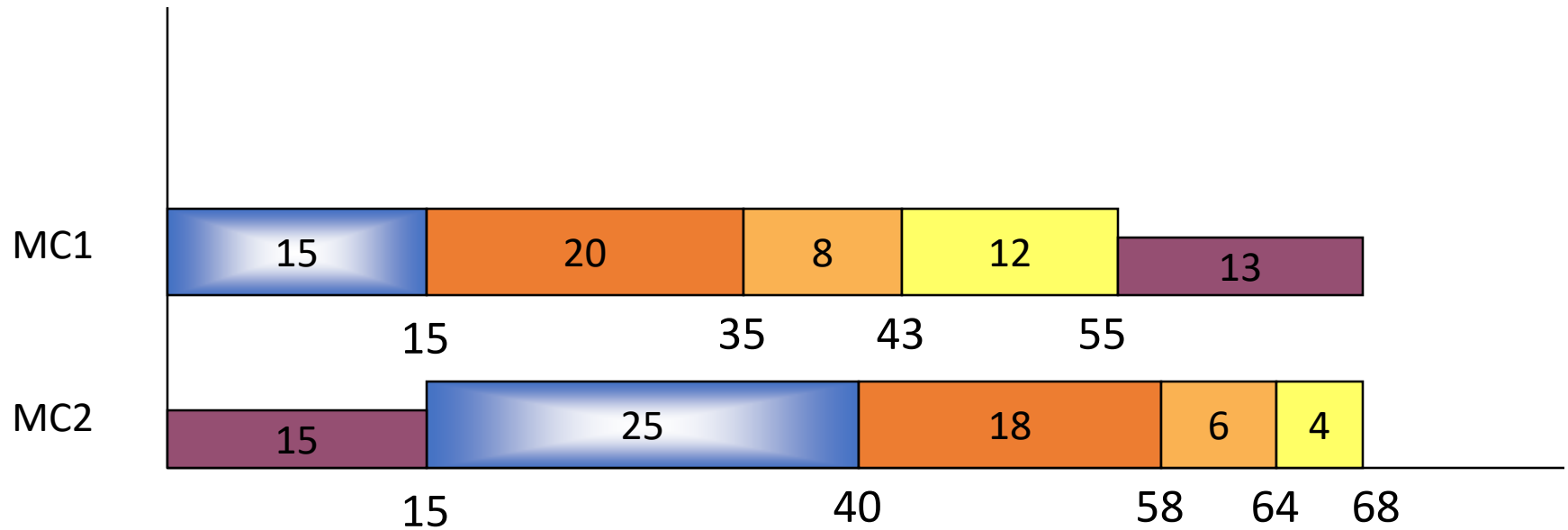
Example of Scenario 2: n Jobs, 2 Machines, flow shop

Job	Processing time on 1	Processing time on 2
A	15	25
B	8	6
C	12	4
D	20	18

- What will be the Optimal Sequence using Johnson rule?
- Develop a Gantt Chart for the schedule & compute the flow times and Make Span for the jobs of this sequence?

The sequence that minimizes the make span

A - D - B - C



Idle time = 28

Makespan = 68

Another Example of
Scenario 2: n Jobs, 2 Machines, flow shop

<i>Job</i>	<i>Machine 1</i>	<i>Machine 2</i>
1	4	7
2	6	3
3	2	3
4	7	7
5	8	6

- What will be the Optimal Sequence using Johnson rule?
- Develop a Gantt Chart for the schedule & compute the flow times and Make Span for the jobs of this sequence?

Optimal sequence : 3 1 4 5 2

Gantt Chart

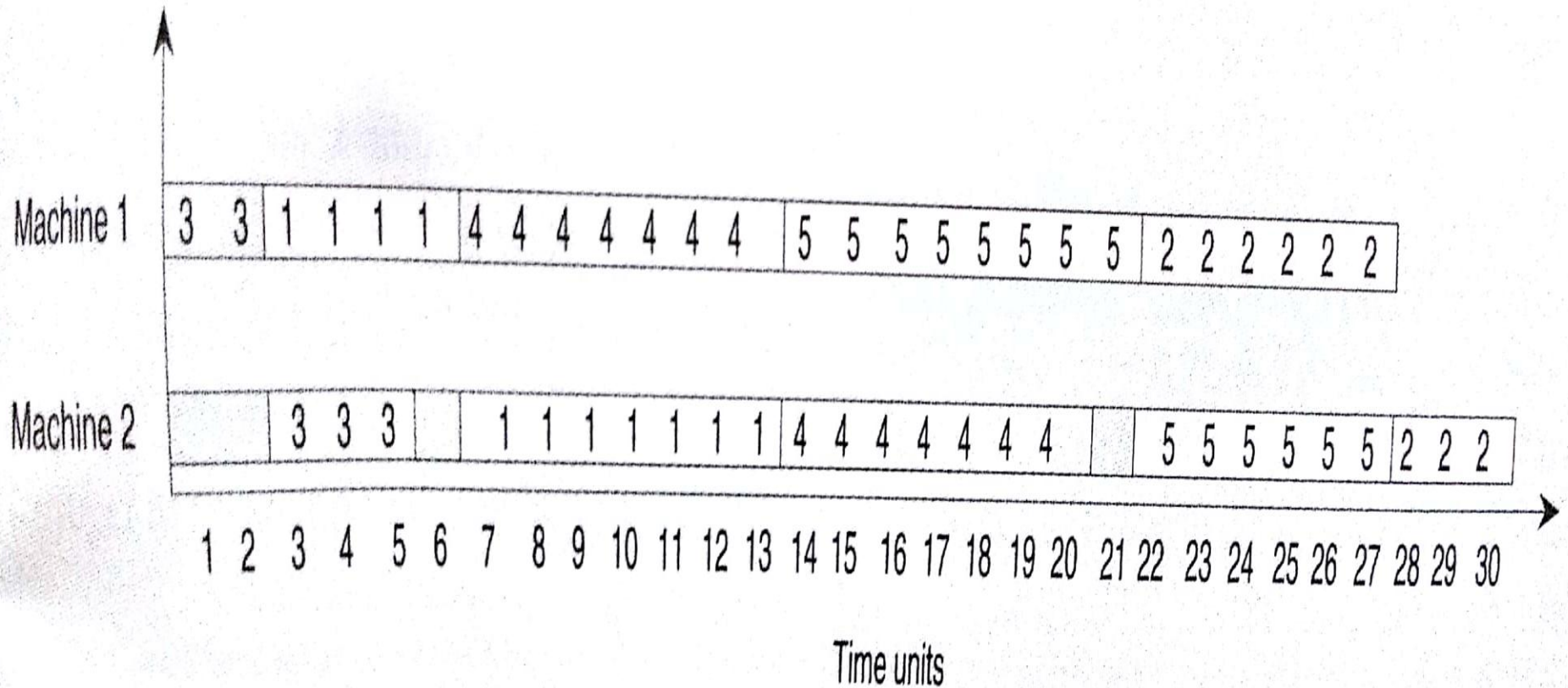


FIGURE 18.4 The resulting Gantt chart

Another Example of Scenario 2: n Jobs, 2 Machines, flow shop

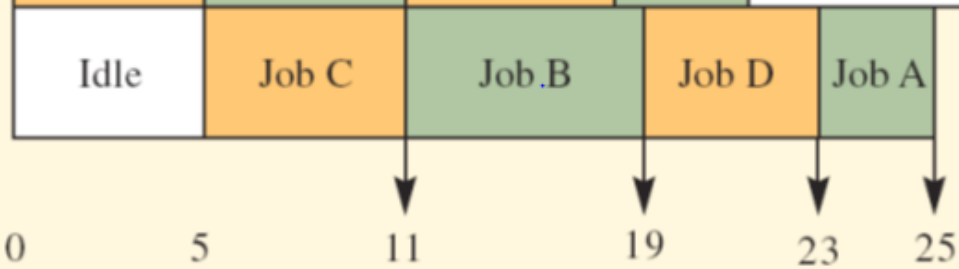
Job	Operation Time on Machine 1	Operation Time on Machine 2
A	3	2
B	6	8
C	5	6
D	7	4

C-B-D-A

Machine 1



Machine 2



0

5

11

19

23

25

Cumulative time in days